

# Dutch Disease or Dutch Blessing?

Shale Gas Shock in the United States and its Impact on Educational Decisions

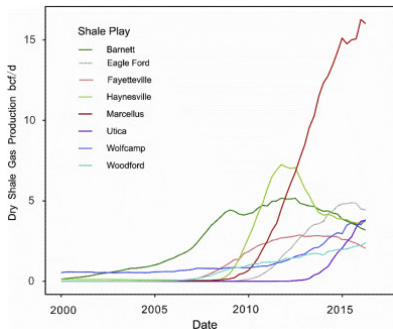
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GitHub Repo: <https://github.com/wooyongp/shale-gas-education>

October 8, 2025

# Is the Shale Gas Boom a Blessing or a Curse?



## Causal Relations

- Shale Gas Boom  $\rightarrow$  Income
- Shale Gas Boom  $\rightarrow$  Education
- HTE by Income Level

Figure: Shale Gas Production per Area in each Well

- Rapid growth in natural resource industries generates numerous well-paying jobs that require relatively little formal education.
- As a result, young people may choose to work instead of staying in school leading to lower educational attainment

- Positive Effects of Shale Gas Shock **on Income**
  - Weinstein and Partridge ([2011](#))
- Negative Effects of Shale Gas Shock **on Education**
  - Rickman, Wang, and Winters ([2017](#)), Carpenter, Anderson, and Dudensing ([2019](#))

## What these papers miss

- No discussions of **heterogeneous treatment effects**, especially by income level
  - Resource booms might increase education for low-income households, but not for high-income households
- They all focus on intra-state district-level comparisons, not cross-state comparisons

# IPUMS ACS Data

- IPUMS ACS - 1yr from 2002 to 2010
- Variables of Interest: Household Income, **College Enrollment**, School Attendance, etc.
- Excluded five states: Hawaii, Alaska, Puerto Rico, Colorado, and Wyoming

|           | Age      | Observations  | Weighted Population | Personal Income   |           |               | Family Income      |           |               |
|-----------|----------|---------------|---------------------|-------------------|-----------|---------------|--------------------|-----------|---------------|
|           |          |               |                     | Mean              | Std. Dev. | Count         | Mean               | Std. Dev. | Count         |
| Non-shale | [5, 10]  | 1,308,465.00  | 170,364,982.00      | NaN               | -0.00     | 0.00          | 72,838.58          | 75,229.40 | 1,298,525.00  |
|           | [16, 18] | 716,276.00    | 89,493,317.00       | 2,152.02          | 5,346.15  | 716,139.00    | 76,209.93          | 75,288.26 | 684,920.00    |
|           | [18, 24] | 1,396,758.00  | 201,384,880.00      | 10,645.98         | 13,378.58 | 1,396,368.00  | 59,542.08          | 63,236.34 | 1,287,939.00  |
|           | total    | 17,029,516.00 | 2,116,225,634.00    | 32,089.57         | 45,426.91 | 13,756,814.00 | 71,707.67          | 72,064.18 | 16,680,293.00 |
| Shale     | [5, 10]  | 301,477.00    | 40,178,779.00       | NaN               | -0.00     | 0.00          | 62,547.86          | 64,017.72 | 299,038.00    |
|           | [16, 18] | 162,607.00    | 20,788,610.00       | 2,107.18          | 5,266.69  | 162,587.00    | 67,489.58          | 66,903.25 | 155,650.00    |
|           | [18, 24] | 315,031.00    | 47,152,003.00       | 10,024.05         | 12,636.98 | 314,959.00    | 52,141.72          | 55,290.96 | 291,304.00    |
|           | total    | 3,794,579.00  | 478,463,294.00      | 28,501.66         | 40,124.54 | 3,043,685.00  | 63,292.86          | 62,940.60 | 3,709,749.00  |
|           | Age      |               |                     | Education (Years) |           |               | College Enrollment |           |               |
|           |          |               |                     | Mean              | Std. Dev. | Count         | Mean               | Std. Dev. | Count         |
| Non-shale | [5, 10]  |               |                     | 0.63              | 1.31      | 1,308,465.00  | 0.00               | 0.00      | 1,308,465.00  |
|           | [16, 18] |               |                     | 10.60             | 1.32      | 716,276.00    | 0.23               | 0.42      | 716,276.00    |
|           | [18, 24] |               |                     | 12.37             | 2.02      | 1,396,758.00  | 0.82               | 0.39      | 1,396,758.00  |
|           | total    |               |                     | 11.14             | 5.28      | 16,429,863.00 | 0.64               | 0.48      | 17,029,516.00 |
| Shale     | [5, 10]  |               |                     | 0.60              | 1.25      | 301,477.00    | 0.00               | 0.00      | 301,477.00    |
|           | [16, 18] |               |                     | 10.48             | 1.40      | 162,607.00    | 0.21               | 0.41      | 162,607.00    |
|           | [18, 24] |               |                     | 12.22             | 2.00      | 315,031.00    | 0.80               | 0.40      | 315,031.00    |
|           | total    |               |                     | 10.74             | 5.21      | 3,656,064.00  | 0.61               | 0.49      | 3,794,579.00  |

Table: Summary Statistics

In our case of shale gas boom, the treatment assignment is staggered across states.

| Year | State         |
|------|---------------|
| 2005 | Texas         |
| 2006 | Arkansas      |
| 2006 | Oklahoma      |
| 2008 | Alabama       |
| 2008 | Louisiana     |
| 2008 | Pennsylvania  |
| 2008 | West Virginia |

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|            | $t_1$ | $t_2$ | $t_3$ | $t_4$ | $t_5$ |
|------------|-------|-------|-------|-------|-------|
| $g_1$      | ○     | ○     | ○     | ○     | ○     |
| $g_2$      | x     | ○     | ○     | ○     | ○     |
| $g_3$      | x     | x     | ○     | ○     | ○     |
| $g_4$      | x     | x     | x     | ○     | ○     |
| $g_5$      | x     | x     | x     | x     | ○     |
| $g_\infty$ | x     | x     | x     | x     | x     |

Table: Shale Gas Extraction Beginnings by State

Table: Staggered Adoption

# Assumptions for Causal identification

- Sun and Abraham ([2021](#)) DiD
  - **Parallel Trends:** The treated and control groups would have followed the same trend in the absence of treatment
  - **No Compositional Changes:** The composition of the treated and control groups remains constant over time
  - **Homogeneity of Dynamic Treatment Effects:** The dynamic evolution of treatment effect is the same across different shale states

# Main Result: Income

- Household income increased by 2.5% overall in the third year after the shale boom.
- In 1Q, the increase was 18%; in 2Q, it was 5%; in 3-4Q, it was not statistically significant.

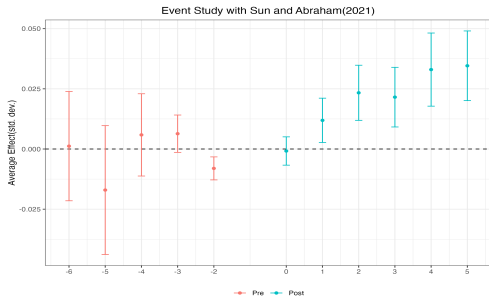


Figure: Overall Income

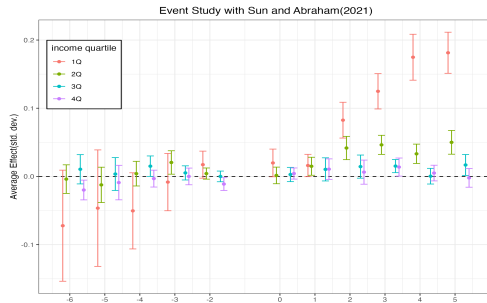


Figure: Income by Quartile

# Main Result: Education

- College enrollment increased by 7.5%p overall in the third year after the shale boom.
- The increase was mainly driven by the 1Q households.
- Drawback: Violation in the parallel trend assumptions in the left figure

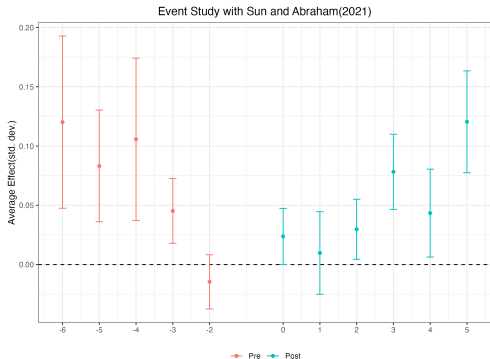


Figure: College Enrollment

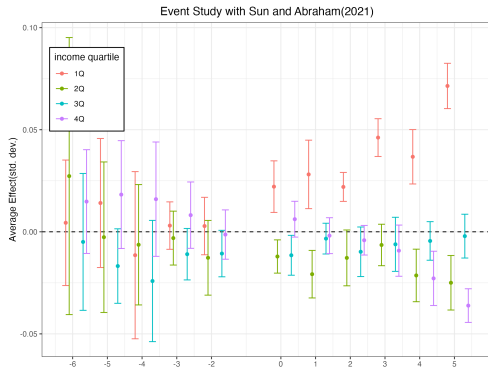


Figure: College Share by Income Quartile



# Future Work

- What kind of microdata can I add to this research?
- How can I incorporate ML methods such as causal forest to this research?
  - Wang, Martinez, and Hahn (2024) proposes the following method to estimate CATT in panel data settings:
  - Assumptions: Unconfoundedness, Overlap
  - Outcome Model

$$Y_{it} = \alpha\mu(X_i, T = t) + \beta_S\nu(X_i, S_{it}, T = t) + \epsilon_{it}$$

where  $S$  is the length to exposure (Train XBART forests separately for  $\mu$  and  $\nu$ )

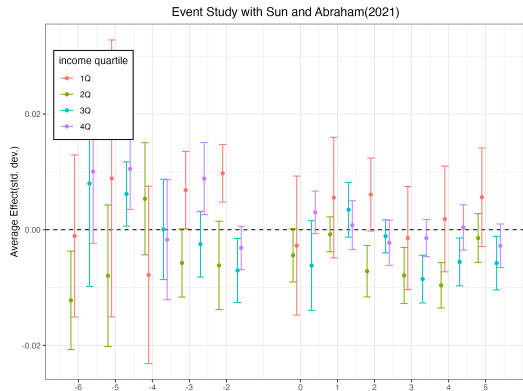
- CATT:

$$\begin{aligned}\tau_t(X_i, S) &\equiv \mathbb{E}[Y_{it}|X_i, S_{it} = S] - \mathbb{E}[Y_{it}|X_i, S_{it} = 0] \\ &= \beta_S\nu(X_i, S_{it}, T = t) - \beta_0\nu(X_i, S = 0, T = t)\end{aligned}$$

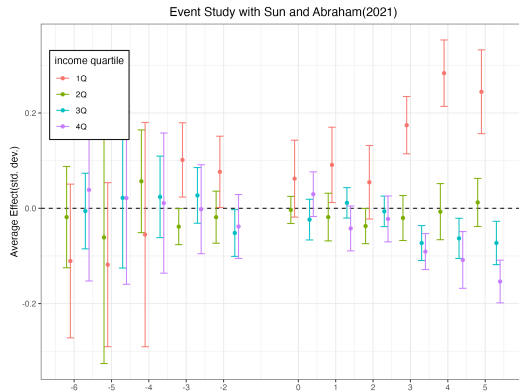
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# Extensions: Other Educational Outcomes



**Figure:** Elementary School Attendance by Income Quartile



**Figure:** High School Attendance by Income Quartile